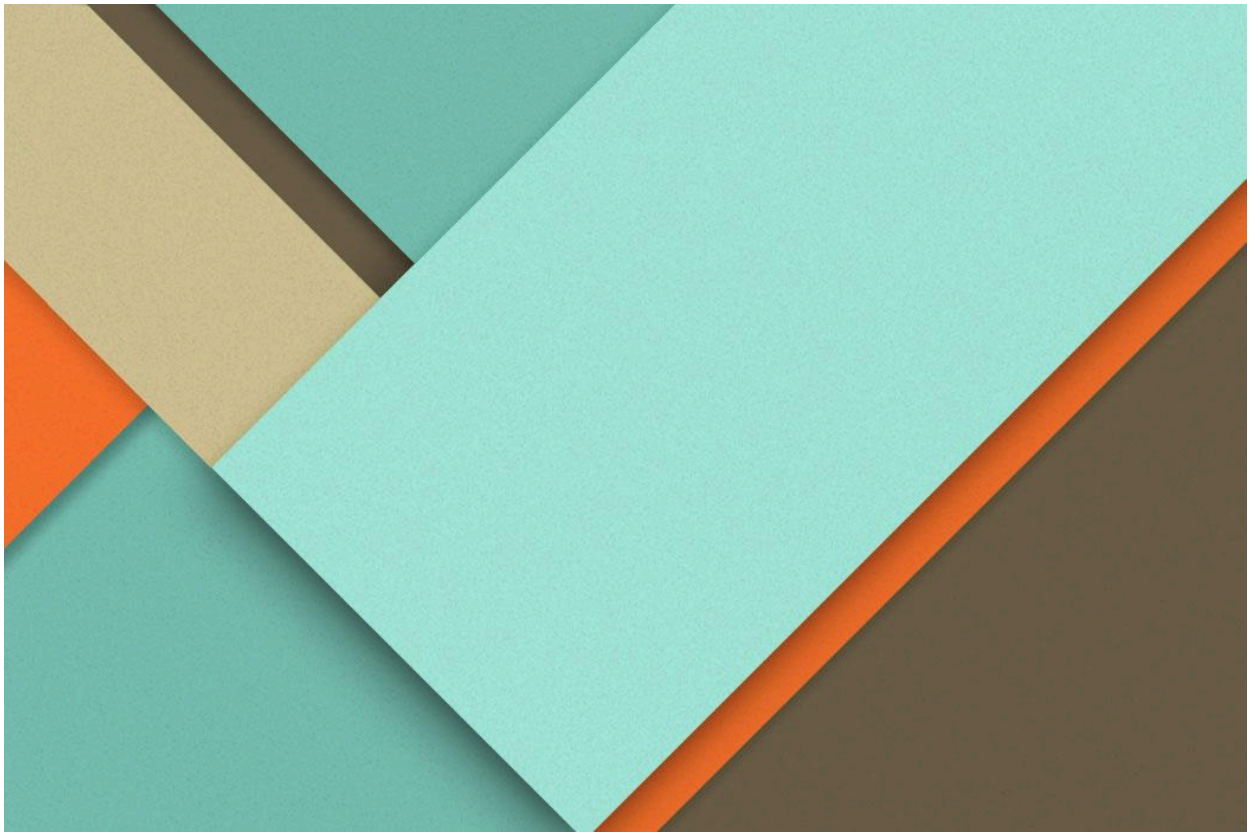

Injury recovery and shoulder health: A training and lifestyle analysis

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Overview

Through the years of wear and tear on my shoulder, I have found myself in a situation where it no longer functions so well. Bicep tendonitis, grade 1 labrum tear, and grade 2 rotator cuff tear. I was previously the captain of my university table tennis team and a coach at a local club.

Goals

1. Reach a comfort level in my shoulder such that I can return to sport and pursue playing at a high level again.
2. Use the information provided by this study to help form a diagnostic process to optimize recovery time for future injuries that can be applied to any area of the body.

Analytic questions

- What lifestyle factors influence my recovery the most?
- What is the best predictor of good recovery, and the best predictor of bad recovery?
- How does overall volume correlate with recovery, and my ability to perform?

Timeline and Methodology

1. I collected data for a month in a google sheet, using my best judgement on basic variables that may affect recovery such as sleep and the amount of stress I am putting on my body on a day to day basis.
2. Upon completion of data, I wrote a python script to clean the data and make some additional variables that may be useful in diagnostic processes
3. With this data I ran regression analysis and exploratory data analysis to try to come to some conclusions about the data. There were a handful of scripts involved that are all in the github repo.
4. Upon completion of all the technical aspects, I wrapped up the project with the following more detailed technical report.

Executive Summary

This study examines the interaction between some daily measurables (such as resting heart rate, sleep duration, volume of weight in a workout) and estimated values (quality of sleep, stress level) to assess how they affect the end goal of rehabbing a nonfunctional shoulder.

A regression based approach was used to determine what values are the most important to a predictive model for predicting the health of my shoulder. The results indicate that volume, sleep quality, and stress were among the best predictors of the dependent variable - shoulder score. It indicates that by having a high stress level you will lower your performance, and that by having increased sleep, you will increase your performance. I imagine these results probably align with what most would have presumed.

Data Description

Shoulder_score (Dependant variable) - This score was guided by a physical therapists grading for shoulder functionality with a 10 being the best score possible and a 0 the worst score possible

Sleep_hours - Hours slept from the prior night

Sleep_quality - An opinion based rating on how well I slept, with 0 being the worst and 10 being the best

Resting_HR - My resting heart rate, taken at a similar time of day every day

Stress_level - An opinion based rating on how stressed I was throughout the day, with 0 being lowest stress and 10 being highest stress

Workout_Class - Categorical variable that just shows the type of workout performed on the day

Volume - Amount of weight moved in the workout, in pounds

Vol_RPE - Amount of weight moved weighed against my rate of perceived exertion in the workout

Sleep_cat - Categorical value that is outlined in greater detail in the data cleaning python script, it determines a more simple category to classify my sleep whether it be poor, neutral, or good.

Sleep_7day_avg - Rolling 7 day average of my sleep hours

RHR_7day_avg - Rolling 7 day average of my resting heart rate

Sleep_lag1 - Lag variable of my sleep hours

RHR_lag1 - Lag variable of my resting heart rate

SS_lag1 - Lag variable of my shoulder score

Results and Analysis

When thinking about this problem logically, some of these results come natural. It is not farfetched to come to the conclusion that poor sleep will give poor physical results, and that being highly stressed will also lead to poor physical results. I think most of this study is fairly intuitive, but it was most interesting to see exactly how interactive or non-interactive these concepts were. The strongest model that I was able to make to predict what my shoulder score would be involved 4 variables: Vol_RPE, Sleep_hours, Stress_level, and Sleep_quality - in that order of significance. It makes sense that a volume based statistic would be the most impactful as naturally a healthy shoulder will be able to handle more volume in a workout. The exact formula is as follows (please see Fig8 for the most indepth look at the regression):

1.2645 + .0087(Vol_RPE) + .2140(Sleep_hours) + .1312(Sleep_quality) - .1281(Stress_level) + error = Shoulder_score

This formula gave the highest observed R-squared value of .261, or 26.1% of the variance in the data is explained by this formula - so still not a very large amount but it has some predictive power. This predictive power is evidenced by the F-stat being 2.291, giving us a p value of .0866. Still not extremely strong, as this would fail an alpha value of 0.05, but it would pass the alpha of .10, so it indicates that there is some moderate strength to this formula, but not something to take as completely law either.

Something I found interesting that I did not investigate deeper due to the already small sample size is outliers. Stress level already seemed to be a strong predictor - strong enough to be fit in the strongest model! But when doing visual analysis I found it to have one of the most striking graphs in comparison to shoulder score, which you can find as (Figure 6.) This graph has one particularly sore outlier, and one can imagine that without that one outlier, this may have been a significantly stronger correlation, and could lead to a more accurate model as a result. If this were a study that had a greater sample size, I would feel more comfortable with deleting entries due to outliers.

Some recommendations I would have for this study in the future would be to try to increase more opinion based variables, as the obvious connection between some of the physical variables feels a bit redundant. It would also perhaps be nice to run this study with a variable that is intentionally controlled such as sleep length (what happens if someone gets 5 hours per day and someone gets 8 hours per day?) Overall it at least confirms some

suspicions that one may have already had and assesses how connected they are so I am happy with the result - I just believe it could be expanded on as well.

Recommendation

This study would lead me to recommend that one practices getting enough sleep, and try to exist in a stable environment if they wish to see progress in their shoulders health. What I mean by this is that the results of this study lead me to believe that if you are in an environment that is not stable (you can not do your rehab exercises enough, you have high variance in stress levels, you have a varying sleep schedule, etc) that your performance will suffer. If you can try to maintain stability in other aspects of your life, physical stability will come easier.

Appendix

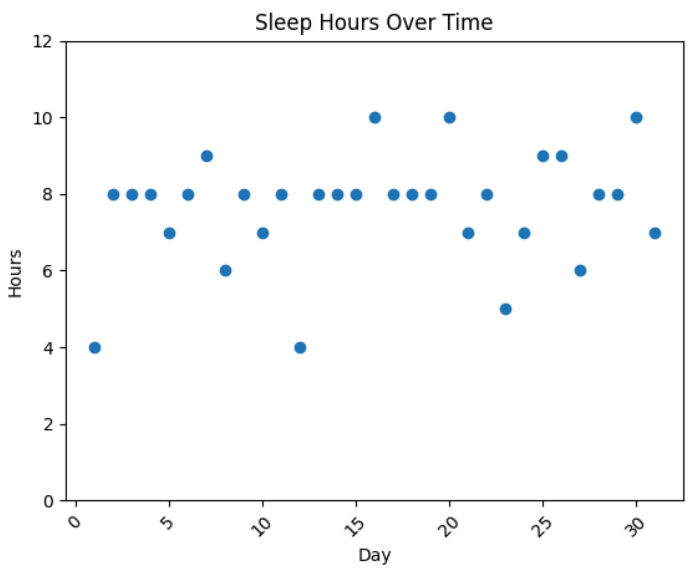

Fig1 - Graph displaying hours slept per night, no significant trends on observation.

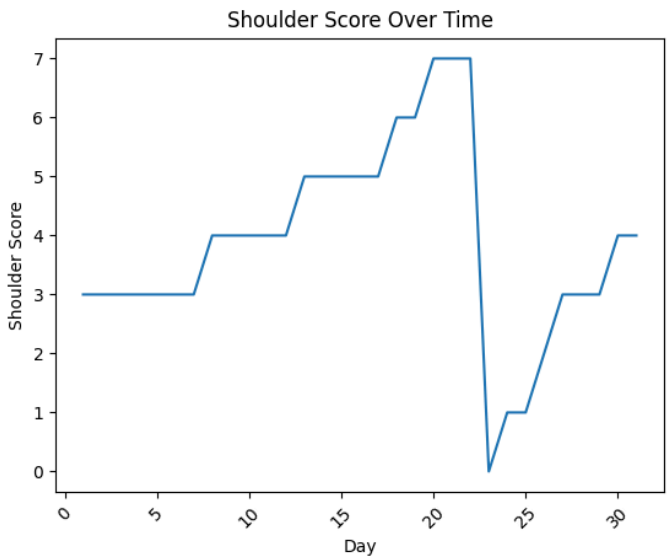

Fig2 - Graph showing shoulder score over time, it essentially continually increased the entire duration of the study, with a large setback around 2/3rd into the study.

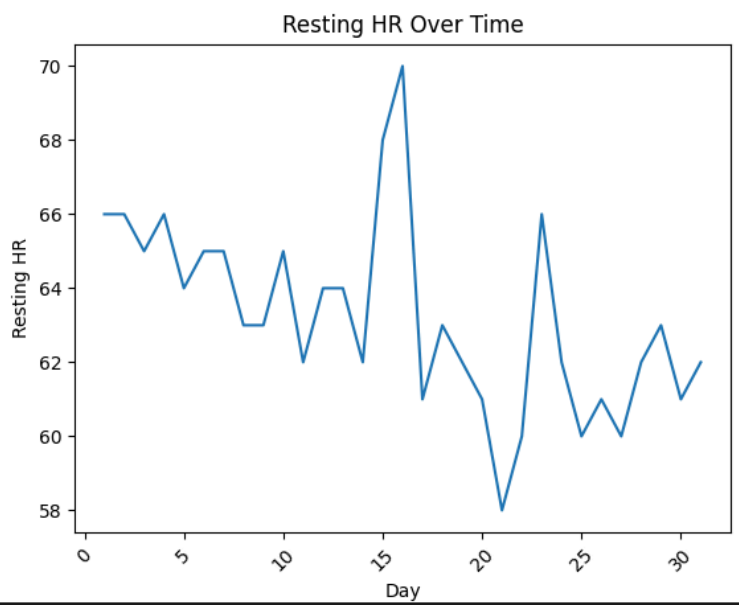

Fig3 - Graph showing resting heart rate over time.

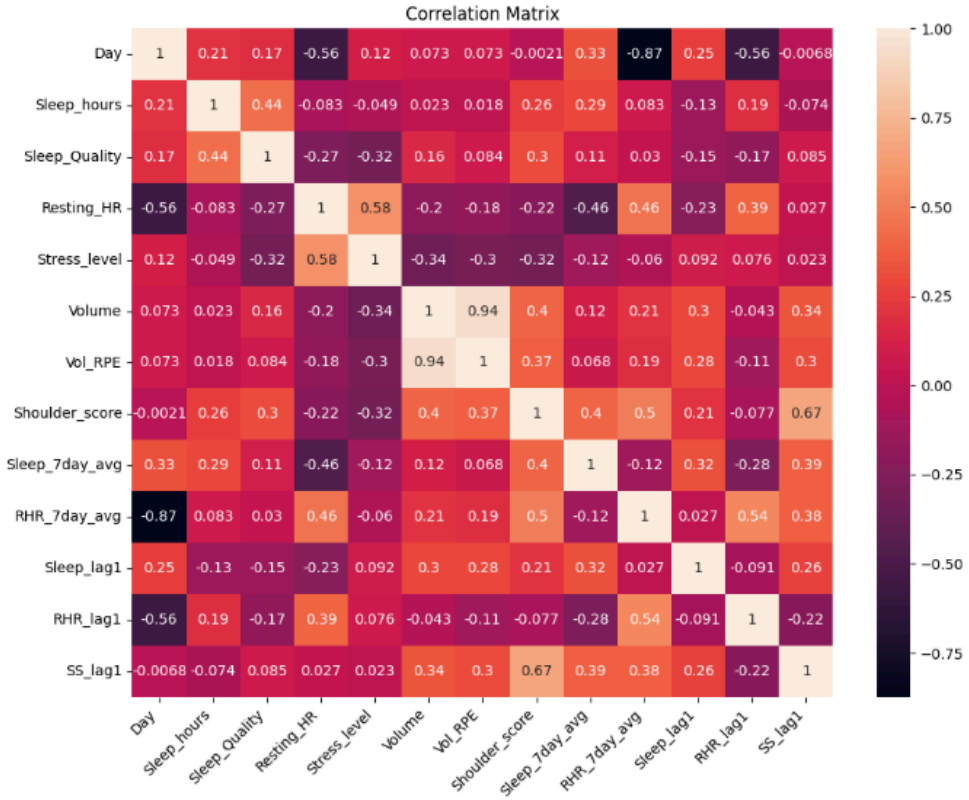

Fig4 - Color scaled correlation matrix of the variables in this study. Some of these entries are not actually worth considering, such as Day vs RHR_7day_avg, as for example the first 7 entries in this are null.

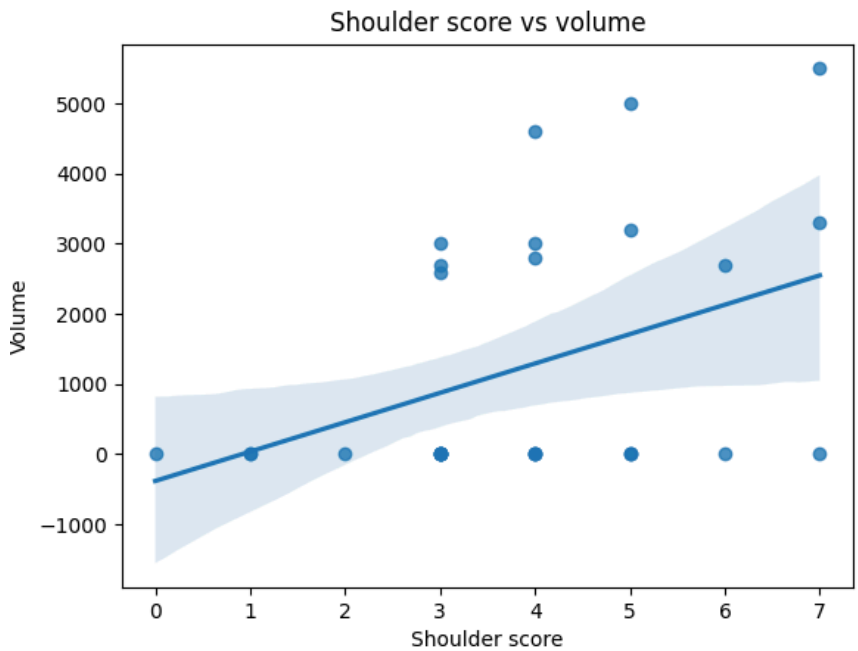

Fig5 - Graph showing shoulder score vs volume, with a best fit line included.

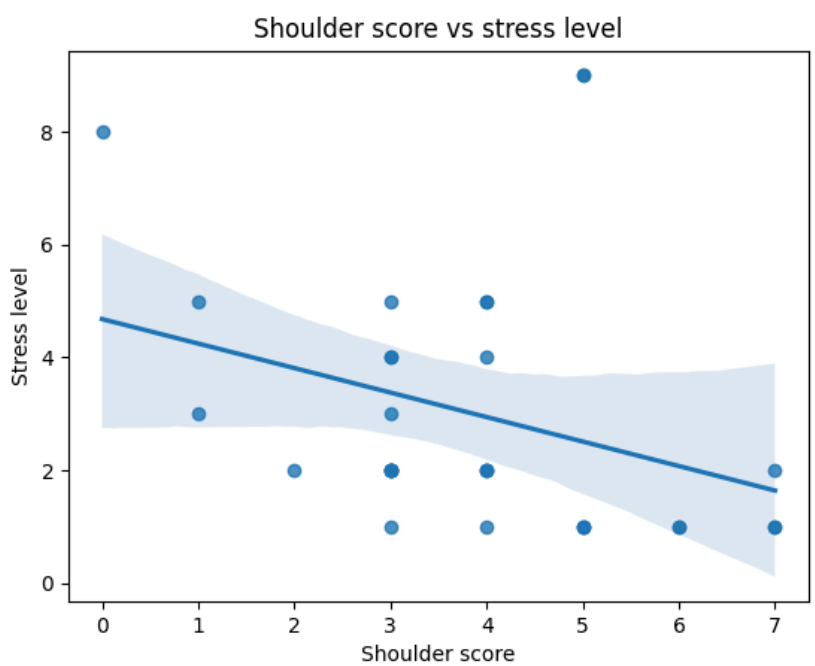

Fig6 - Shoulder score vs stress level, with a best of fit line included. As previously mentioned, a visual that I think is important to the study in showing how important this variable really is.

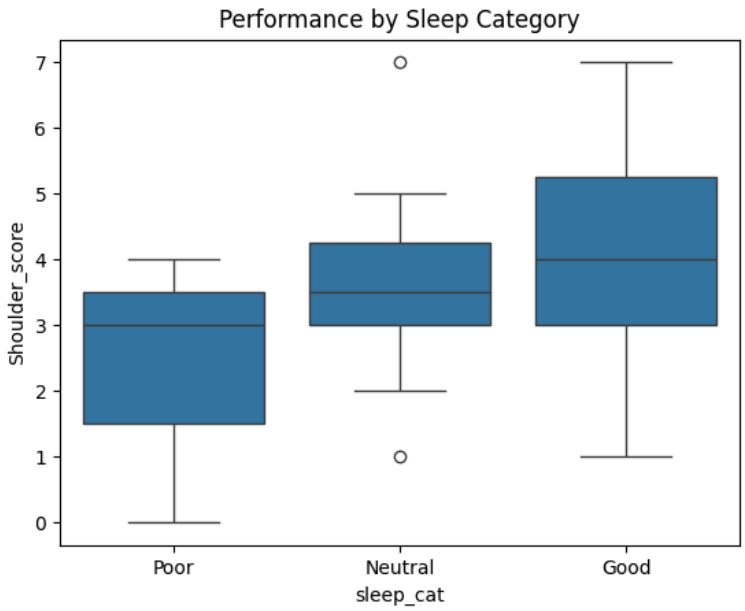

Fig7 - A box and whisker plot that I really enjoy - we can observe that the good category has greater variance but still the best average shoulder performance. This is likely because the good category is also the category I most often fell into for my sleep.

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                            OLS Regression Results
==============================================================================
Dep. Variable:         Shoulder_score   R-squared:                       0.261
Model:                            OLS   Adj. R-squared:                  0.147
Method:                 Least Squares   F-statistic:                     2.291
Date:                Tue, 31 Mar 2026   Prob (F-statistic):             0.0866
Time:                        20:11:27   Log-Likelihood:                -55.387
No. Observations:                  31   AIC:                             120.8
Df Residuals:                      26   BIC:                             127.9
Df Model:                           4
Covariance Type:            nonrobust
==================================================================================
                     coef    std err          t      P>|t|      [0.025      0.975]
----------------------------------------------------------------------------------
const              1.2645      1.801      0.702      0.489      -2.437       4.966
Sleep_hours        0.2140      0.219      0.978      0.337      -0.236       0.664
Sleep_Quality      0.1312      0.188      0.697      0.492      -0.256       0.518
Stress_level      -0.1281      0.137     -0.934      0.359      -0.410       0.154
Vol_RPE            0.0087      0.005      1.742      0.093      -0.002       0.019
==============================================================================
Omnibus:                        0.427   Durbin-Watson:                   0.824
Prob(Omnibus):                  0.808   Jarque-Bera (JB):                0.573
Skew:                           0.138   Prob(JB):                        0.751
Kurtosis:                       2.394   Cond. No.                         473.
==============================================================================
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Fig8 - Regression analysis results. We can observe the R-squared explaining 26.1% of variance and the F-stat clearing the study on an alpha of .1 but not .05. One can also observe the importance of each of the 4 variables to the model based on their P > t score.